

# Uncountably many commutant classes of extreme points in a $D$ -majorization set

Candidate counterexample for arXiv:2004.05613

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**Status: candidate counterexample/full negative answer, likely valid.** For every nonscalar full-rank density matrix  $D$ , a pure state  $A$  can be chosen so that  $M_D(A)$  is the entire state space. Modulo unitaries commuting with  $D$ , its extreme points form a positive-dimensional simplex and hence have uncountably many equivalence classes.

## 1 Source and open question

The source is Frederik vom Ende, *Strict Positivity and  $D$ -Majorization*, arXiv:2004.05613, subsequently published in *Linear and Multilinear Algebra* 70:19 (2022), 4023–4048 [1]. For a positive definite matrix  $D$ , let

$$Q_D(n) = \{T : M_n(\mathbb{C}) \rightarrow M_n(\mathbb{C}) : T \text{ is completely positive and trace preserving, } T(D) = D\}.$$

The source writes  $X \prec_D A$  when  $X = T(A)$  for some  $T \in Q_D(n)$ , and

$$M_D(A) = \{T(A) : T \in Q_D(n)\}.$$

In Section 5, pages 20–21, the source observes that conjugation by a unitary commuting with  $D$  preserves the extreme points of  $M_D(A)$ . It then asks whether factoring out this unitary equivalence leaves only finitely many extreme points (meaning finitely many equivalence classes).

## 2 Proof intuition

Choose  $A$  to be a pure state in a minimum-eigenvalue eigenspace of  $D$ . For every desired output state  $\rho$ , the minimum-eigenvalue inequality makes  $D - d\rho$  positive. This produces an explicit measure-and-prepare channel that fixes  $D$  and sends  $A$  to  $\rho$ . Thus  $M_D(A)$  is not merely a complicated convex body: it is the whole state space.

Its extreme points are the pure states. A unitary commuting with  $D$  acts independently inside the distinct spectral subspaces of  $D$ , so its orbit remembers exactly how much squared norm a pure vector has in each such subspace. These weights fill a simplex. When  $D$  is nonscalar the simplex has positive dimension, giving uncountably many inequivalent extreme points.

## 3 Counterexample and exact quotient

Write  $\mathcal{D}_n = \{\rho \in M_n(\mathbb{C}) : \rho \succeq 0, \text{ tr } \rho = 1\}$ .

0.0 (17), or Remark 2 depending on the physical picture, as a definition for finite dimensional systems as then one could choose whether the orthogonal projections  $\pi$  have to be finite-dimensional or can be arbitrary.

Either way strict positivity proves to be a useful tool in analyzing  $D$ -majorization on matrices and its order properties. The biggest difference between the vector and the general matrix case arguably lies within the convex compact set of all matrices  $D$ -majorized by some initial matrix from Eq. (9): in the vector case this set has finitely many extreme points meaning it forms a convex polytope. This grants access to strong tools from the underlying mathematical theory and allows for an analytical extreme point analysis. In the matrix case, however, the number of extreme points is infinite<sup>12</sup>—

<sup>11</sup> For bounded sequences  $(A_n)_{n \in \mathbb{N}}$  of non-empty compact subsets (of some metric space) which Hausdorff-converges to  $A$ , one has  $x \in A$  if and only if there exists a sequence  $(a_n)_{n \in \mathbb{N}}$  with  $a_n \in A_n$  which converges to  $x$ , cf. [53, Lemma 2.4].

<sup>12</sup> Similarly while the set of  $d$ -stochastic matrices forms a convex polytope, the set  $Q_D(n)$  has infinitely many extreme points. The argument, just as below, relies on the concatenation with suitable unitary channels from

a straightforward argument shows that if  $X$  is extremal in  $M_D(A)$  then so is  $U^*XU$  for all unitaries  $U$  which satisfy  $[U, D] = 0$ . A way of bridging this gap could be to define the unitary equivalence relation

$$\sim := \{(X, UXU^*) \mid X \in M_D(A), U \in \mathbb{C}^{n \times n} \text{ unitary with } [U, D] = 0\}$$

for arbitrary  $d \in \mathbb{R}_{++}^n$  and  $A \in \mathbb{C}^{n \times n}$ , and then look at the equivalence class of an extreme point  $X$  of  $M_D(A)$  under  $\sim$ . Whether one ends up with finitely many extreme points after factoring out the unitary equivalents for now remains open.

Other than simplifying extreme point analysis answering this question might also enable a proof for continuity of the map  $(D, P) \mapsto M_D(P)$  defined for  $D > 0$  and  $P \in \mathcal{P}_c(\mathbb{C}^{n \times n})$ —the latter possibly restricted to just the hermitian matrices. In the vector case this map is indeed continuous [18, Thm. 4.8] but the proof relies heavily on the polytope structure and the corresponding half-space representation.

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Figure 1: The concluding discussion and open finiteness question on pages 20–21 of the source PDF.

**Theorem 1.** *Let  $D \in \mathcal{D}_n$  be positive definite and nonscalar, with  $n \geq 2$ . Choose a unit eigenvector  $e$  for  $d = \lambda_{\min}(D)$ , and put  $A = |e\rangle\langle e|$ . Then*

$$M_D(A) = \mathcal{D}_n.$$

*If  $D = \sum_{j=1}^r \lambda_j P_j$  is its spectral decomposition over the distinct eigenvalues, then the extreme points of  $M_D(A)$ , modulo conjugation by unitaries commuting with  $D$ , are in bijection with*

$$\Delta_{r-1} = \left\{ (w_1, \dots, w_r) : w_j \geq 0, \sum_{j=1}^r w_j = 1 \right\}.$$

*In particular, there are uncountably many equivalence classes.*

*Proof.* Fix an arbitrary state  $\rho \in \mathcal{D}_n$ . Since  $D \succeq dI$  and every density matrix satisfies  $\rho \preceq I$ ,

$$D - d\rho = (D - dI) + d(I - \rho) \succeq 0. \tag{1}$$

Also  $0 < d < 1$ , and hence

$$\tau := \frac{D - d\rho}{1 - d}$$

is a density matrix.

Extend  $e = e_1$  to an orthonormal eigenbasis  $e_1, \dots, e_n$  of  $D$ , and define

$$T_\rho(X) = \langle e_1, X e_1 \rangle \rho + \sum_{k=2}^n \langle e_k, X e_k \rangle \tau. \quad (2)$$

This is a measure-and-prepare channel, hence completely positive. For completeness, diagonalizing each prepared state produces Kraus operators  $\sqrt{s_{k\ell}} |v_{k\ell}\rangle \langle e_k|$ ; their adjoint products sum to  $I$ , so  $T_\rho$  is trace preserving. Moreover,

$$T_\rho(A) = \rho, \quad T_\rho(D) = d\rho + (1 - d)\tau = D.$$

Thus  $\rho \in M_D(A)$ . Since a channel maps a state to a state, the reverse inclusion  $M_D(A) \subseteq \mathcal{D}_n$  is automatic. This proves  $M_D(A) = \mathcal{D}_n$ .

The extreme points of  $\mathcal{D}_n$  are exactly the rank-one projections. Indeed, a state of rank at least two can be perturbed by opposite small traceless diagonal perturbations inside its support, while any convex decomposition of a rank-one state forces both summands to have the same one-dimensional support.

Represent an extreme point as  $E_\psi = |\psi\rangle\langle\psi|$ , with  $\|\psi\| = 1$ , and define its spectral weight vector

$$w(\psi) = (\|P_1\psi\|^2, \dots, \|P_r\psi\|^2) \in \Delta_{r-1}.$$

If  $U$  commutes with  $D$ , then  $U$  preserves each spectral subspace, so  $w(U\psi) = w(\psi)$ . Conversely, suppose  $w(\psi) = w(\varphi)$ . On each  $P_j\mathbb{C}^n$ , choose a unitary  $U_j$  sending  $P_j\psi$  to  $P_j\varphi$  (with an arbitrary choice when both vectors vanish). Then  $U = \bigoplus_j U_j$  commutes with  $D$  and satisfies  $U\psi = \varphi$ . Hence  $E_\psi$  and  $E_\varphi$  are equivalent if and only if their weight vectors agree.

Every point of  $\Delta_{r-1}$  is realized by choosing unit vectors  $u_j \in P_j\mathbb{C}^n$  and setting  $\psi = \sum_j \sqrt{w_j} u_j$ . The quotient is therefore exactly  $\Delta_{r-1}$ . Nonscalarly gives  $r \geq 2$ , so this simplex, and hence the set of equivalence classes, is uncountable.  $\square$

**Corollary 1** (Explicit qubit counterexample). *Let*

$$D = \begin{pmatrix} 1/3 & 0 \\ 0 & 2/3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

*Then  $M_D(A) = \mathcal{D}_2$ , and the extreme points*

$$\left| \sqrt{t} e_1 + \sqrt{1-t} e_2 \right\rangle \left\langle \sqrt{t} e_1 + \sqrt{1-t} e_2 \right|, \quad 0 \leq t \leq 1,$$

*are pairwise inequivalent under unitaries commuting with  $D$ .*

*Proof.* The equality follows from Theorem 1. Because  $D$  has simple spectrum, its commuting unitaries are diagonal and preserve the parameter  $t = |\langle e_1, \psi \rangle|^2$ . Distinct parameters are therefore inequivalent.  $\square$

## 4 Verification, scope, and novelty

The proof is exact and has no computational dependency. The main hypothesis checks are equation (1), complete positivity and trace preservation of (2), and the converse half of the spectral weight classification. All are written explicitly above and audited in the companion `verification.md`.

The theorem gives a full negative answer to general finiteness as the source question is written. It does not classify all pairs  $(D, A)$ , and in particular it leaves open whether useful restricted classes of nonmaximal initial states have finite quotients. For scalar  $D$ , the commutant is the full unitary group and the quotient in this maximal example has one class; nonscalarity is the exact threshold for the construction.

Before promotion, the run indexes were searched using arXiv:2004.05613 and the terms  $D$ -majorization,  $M_D(A)$ , commutant unitaries, unitary equivalence, and extreme points. A bounded web/arXiv search on 11 August 2026 used the exact open sentence, close variants, and citation/title searches. It found the source, the author's 2020 dissertation repeating the question, the classical  $d$ -majorization polytope literature, and unrelated majorization-orbit results, but no stated resolution.

Novelty confidence is moderate rather than high. The explicit channel is the same mechanism underlying the source's maximal-state theorem, but the uncountable quotient classification and its negative consequence were not found in the bounded search. Human review is recommended, with special attention to whether the informal conclusion intended to exclude maximal initial states; the printed question contains no such restriction.

## References

- [1] F. vom Ende, *Strict Positivity and  $D$ -Majorization*, Linear Multilinear Algebra 70:19 (2022), 4023–4048; arXiv:2004.05613. <https://arxiv.org/abs/2004.05613>.